

Exercise 05

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Problem 1.

Solution. 1. F - $C = \{\{a\}, \{b, c\}, \{d, e\}\}$

2. F

3. T

4. F

5. T - $\mathbb{Z}$ such that $p_{x,x} = 1$ for $x \in \mathbb{Z}$

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Problem 1b.

Solution. 1. F - If $\epsilon = 0$, $C = \{\{a, b\}, \{c, d\}\}$

2. F - Assume π is reversible then $\pi_c p_{c,d} = \pi_d p_{d,c} \implies \frac{1}{4} p_{c,d} = \frac{1}{4} p_{d,c} \implies p_{c,d} = \frac{2}{3} = \frac{1}{3} = p_{d,c}$ which is a contradiction.

3. T -

$$(a) \ a + b: \pi_a p_{a,b} = \frac{1}{2} * \frac{1}{8} = \frac{1}{2} * \frac{1}{8} = \pi_b p_{b,a}$$

$$(b) \ c + d: \pi_c p_{c,d} = \frac{2}{3} \frac{1}{4} = \frac{1}{6} = \frac{1}{2} \frac{1}{3} = \pi_d p_{d,c}$$

$$(c) \ b + c: \pi_b p_{b,c} = \frac{1}{8} \epsilon = \frac{1}{4} \epsilon = \pi_c p_{c,b} \iff \epsilon = 2\epsilon \iff \epsilon = 0.$$

Thus π is reversible $\iff \epsilon = 0$

4. T - for $\epsilon = 0$ π is reversible. Since reversible implies stationary, π is stationary for $\epsilon = 0$

5. Let P_ϵ be the transition matrix for the markov chain. $\pi P_\epsilon = \pi \ \forall \epsilon \in [0, \frac{1}{3}]$. Thus π is stationary for all $\epsilon \in [0, \frac{1}{3}]$

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Problem 2a.

Proof. 1. $\forall x_0, x_1 \in \mathbb{N} \cup \{0\}$, $\mathbb{P}(X_0 = x_0) = \mathbb{P}(X_0 = x_0)\mathbb{P}(X_1 = x_1 \mid X_0 = x_0) = \delta_{x_0}^1 p_{x_0, x_1}$.

Assume that $\forall x_0, \dots, x_n \in \mathbb{N} \cup \{0\}$

$$\mathbb{P}(X_n = x_n, \dots, X_0 = x_0) = \delta_{x_0}^1 \prod_{i=1}^n p_{x_{i-1}, x_i}$$

$\forall x_0, \dots, x_{n+1} \in \mathbb{N} \cup \{0\}$

$$\begin{aligned} \mathbb{P}(X_{n+1} = x_{n+1}, X_n = x_n, \dots, X_0 = x_0) &= \mathbb{P}(X_n = x_n, \dots, X_0 = x_0)\mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n, \dots, X_0 = x_0) \\ &= \delta_{x_0}^1 \prod_{i=1}^n p_{x_{i-1}, x_i} \mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n) \text{ (definition of process)} \\ &= \delta_{x_0}^1 \prod_{i=1}^{n+1} p_{x_{i-1}, x_i} \end{aligned}$$

Thus $(X_n)_{n \geq 0} \sim MC(\delta^1, P)$.

2. $\forall (x, y) \in \mathbb{N} \times (\mathbb{N} \cup \{0\})$,

$$\begin{aligned} p_{x,y} &:= \sum_{n_1, \dots, n_x \in \mathbb{N} \ni \sum n_i = y} \prod_{i=1}^x \nu(n_i) \\ p_{0,y} &:= \delta_y^0 \end{aligned}$$

□

Problem 2b.

Proof. Note $\forall x \in \mathbb{N}$, $p_{0,x} = 0$. Thus 0 only communicates with itself. $\{0\}$ is closed.

$\forall x, y \in \mathbb{N} \ni y \leq x$. We want to show that $x \rightarrow y$. Then we want to show that $1 \rightarrow x$.

$$p_{x,y} \geq \nu(0)^{x-y} \nu(1)^y > 0$$

The last inequality comes from the fact that $\nu(0), \nu(1) > 0$.

Thus $x \rightarrow y$.

Now for $1 \rightarrow x$. Note that $\nu(0) + \nu(1) < 1$, thus $\exists n > 1 \ni \nu(n) > 0$. $\exists k \in \mathbb{N} \ni x \leq kn$. We want to show $1 \rightarrow kn$ is true.

$$p_{1,kn}^{(k)} \geq p_{1,n} \prod_{i=2}^k p_{(i-1)n, in} \geq \nu(n) \prod_{i=2}^k \nu(0)^{(i-1)n-i} \nu(n)^i > 0$$

Then $kn \rightarrow x$. Thus $1 \rightarrow x$. Thus x communicates with 1. Since communication is an equivalence relation $\mathbb{N}$ is communication class. Note $p_{x,0} = \nu(0)^x > 0 \implies x \rightarrow 0$. Thus $\mathbb{N}$ is not closed. □

Problem 2c.

Proof. $\mathbb{N}$ is transient since it is not closed.

Note for $n \in \mathbb{N}$ if $X_n = 0$, then $X_{n+1} = 0$ a.s. Thus $P_0(V_0 = \infty) = 1$. Thus $V_0 = \infty$ a.s. Thus $\{0\}$ is recurrent. $\square$

Problem 2d.

Proof. if $\nu(0) = 0$ and $\nu(1) < 1$. That means $\forall x, y \in \mathbb{N} \ni y < x \implies y \rightarrow x$ but $x \not\rightarrow y$. Thus $\{x\}$ is a communication class. Thus $\mathbb{N} \cup \{0\}$ are the set of communication classes.

Only $\{0\}$ is closed.

(c) doesn't change. $\square$

Problem 3a.

Proof. $S = [0, \dots, N]$ where the following transition probability is followed, $\forall x, y \in [0, \dots, N]$

$$p_{x,y} = \begin{cases} \frac{i}{N} & \text{if } y = x - 1 \\ \frac{N-i}{N} & \text{if } y = x + 1 \\ 0 & \text{otherwise} \end{cases}$$

$\square$

Problem 3b.

Proof.

$$\pi = \left(\binom{N}{0} 2^{-N}, \binom{N}{1} 2^{-N}, \dots, \binom{N}{N} 2^{-N} \right)$$

is a reversible distribution of the model.

Suppose $\pi = (\pi_0, \dots, \pi_N)$ is a reversible distribution for the model. $\forall i \in [0, \dots, N-1]$ it follows the following rule:

$$\pi_i p_{i,i+1} = \pi_{i+1} p_{i+1,i}$$

That means

$$\begin{aligned} \pi_i \frac{N-i}{N} &= \pi_{i+1} \frac{i+1}{N} \\ \pi_{i+1} &= \frac{N-i}{i+1} \pi_i \end{aligned}$$

Thus

$$\begin{aligned}
\pi_1 &= N\pi_0 \\
\pi_2 &= N\left(\frac{N-1}{2}\right)\pi_0 \\
&\vdots \\
\pi_m &= \pi_0 \prod_{i=0}^{m-1} \left(\frac{N-1}{i+1}\right) \\
&= \pi_0 \left(\frac{N}{1}\right) \left(\frac{N-1}{2}\right) \cdots \left(\frac{N-m+1}{m}\right) \\
&= \frac{\pi_0}{m!} \left(\frac{N!}{(N-m)!}\right) \\
&= \binom{N}{m} \pi_0
\end{aligned}$$

Then

$$\begin{aligned}
1 &= \sum_{i=0}^N \pi_i \\
&= \pi_0 \left(1 + \sum_{i=1}^N \binom{N}{i}\right) \\
&= \pi_0 \sum_{i=0}^N \binom{N}{i} \\
&= 2^N \pi_0 \implies \pi_0 = 2^{-N} \implies \pi_m = \binom{N}{m} 2^{-N}
\end{aligned}$$

Thus

$$\pi = \left(\binom{N}{0} 2^{-N}, \binom{N}{1} 2^{-N}, \dots, \binom{N}{N} 2^{-N}\right)$$

is a reversible thus stationary distribution. □

Problem 4.

Proof. Note

$$P = \begin{bmatrix} p & 1-p \\ q & 1-q \end{bmatrix}$$

One direction is trivial. If π is reversible it is stationary.

$\implies$: Suppose π is stationary. Thus

$$\pi P = \pi \implies p\pi_1 + (1-q)\pi_2 = \pi_1$$

Thus subtracting $p\pi_1$ from both sides of equation 1 gives us:

$$p_{2,1}\pi_2 = (1 - q)\pi_2 = (1 - p)\pi_1 = p_{1,2}\pi_1$$

Thus π is reversible. □

Problem 5a.

Proof. □

Problem 5b.

Proof. Note the following,

$$\begin{aligned}\langle Pf, g \rangle_\pi &= \sum_{x \in S} (Pf)(x)g(x)\pi_x \\ &= \sum_{x \in S} \sum_{y \in S} p_{x,y}f(y)g(x)\pi_x \\ &= \sum_{y \in S} \sum_{x \in S} p_{x,y}f(y)g(x)\pi_x\end{aligned}$$

□